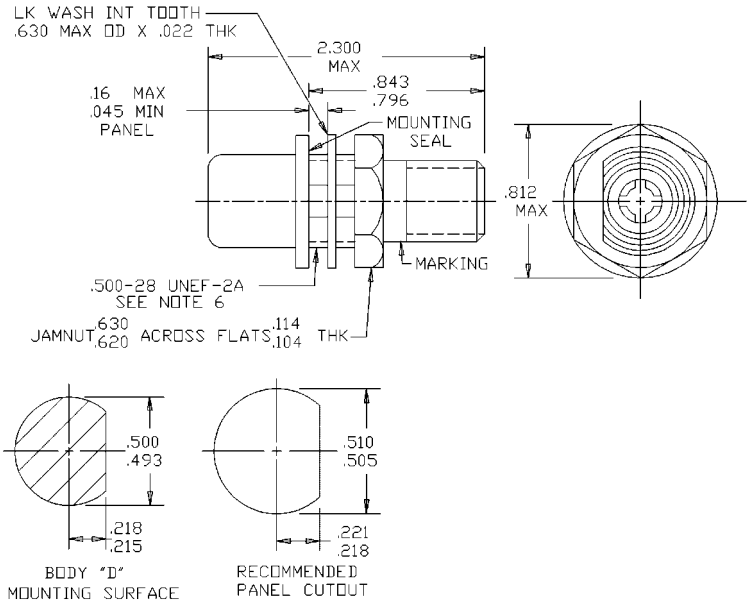
INCH-POUND

MIL-PRF-49142/9C
27 February 2002
SUPERSEDING
MIL-PRF-49142/9B
5 November 1990

PERFORMANCE SPECIFICATION SHEET

CONNECTOR, TRIAXIAL, RADIO REQUENCY, (SERIES TRT (CABLED) RECEPTACLE, SOCKET CONTACT, JAMNUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.



Inches	mm	Inches	mm
.022	0.56	.500	12.70
.045	1.14	.505	12.83
.104	2.64	.510	12.95
.114	2.90	.620	15.75
.16	4.1	.630	16.00
.215	5.46	.796	20.22
.218	5.54	.812	20.62
.221	5.61	.843	21.14
.493	12.52	2.300	58.42

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28.
4. 2.300 (58.42 mm) defines the maximum length of the connector when assembled to the appropriate cable.
5. All undimensioned pictorial representations are for reference purpose only.
6. Full threads to within .063 (1.60 mm) of shoulder.
7. Interface shall be in accordance with MIL-STD-348, series TRT, socket contact.

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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ENGINEERING DATA:

Nominal impedance: Nonconstant.

Frequency range: 0 to 500 MHz minimum.

Voltage rating: 400 V rms maximum working voltage at sea level. 100 V rms maximum working voltage at 70,000 feet.

Temperature range: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2.5 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque: 4 to 6 inch-pounds.

Mating characteristics: See MIL-STD-348 for dimensions.

Center contact (socket):

Oversize test pin: 0.040 inch diameter minimum (nonclosed entry contacts only).

Insertion depth: .125 inch minimum.

Number of insertions: 1.

Insertion force test:

Steel test pin finish: 16 microinches.

Insertion force: 2 pounds maximum.

Steel test pin diameter: .039 inch minimum.

Withdrawal force test:

Steel test pin diameter: .037 inch maximum.

Withdrawal force: 2 ounces minimum.

Test pin finish: 16 microinches.

Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: Not applicable.

Insulation resistance: 5,000 megohms.

Center conductor retention: 6 pounds minimum axial force.

Dielectric withstanding voltage: At sea level, 1,200 V rms, between center conductor and intermediate conductor; 500 V rms, between intermediate conductor and outer conductor.

Salt spray (corrosion): Applicable.

Vibration: Applicable.

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Shock: Applicable.

Thermal shock: Applicable (except high test temperature shall be +200°C for connectors using +200°C cables).

Moisture resistance: Applicable.

Conductor resistance: In milliohms, maximum.

	Initial	After environment
Center conductor:	2.0	2.5
Intermediate conductor	0.5	0.6
Outer conductor	0.2	0.3

Dash number and applicable cable: See table I.

TABLE I. Dash number and applicable cable.

Dash number	Cable 1/
Category A – No special tools required 2/ 3/	
0001	M17/134-00001 M17/134-00003
0002	M17/134-00002 M17/134-00004
0003 4/	D3-7619-5/336
0004 4/	D3-7619-5/338
0005 4/	380-10045-1
0006 3006	M17/176-00002 5/
0007	M17/116-RG307
0009	M17/177-00001 5/
0010	M17/178-00001
0011	M17/179-00001
0012 3012	M24643/33-01UN
0022	M17/135-00003 M17/135-00005
0023	M17/135-00004 M17/135-00006

See footnotes at end of table.

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TABLE I. Dash number and applicable cable – Continued.

Dash number	Cable 1/
Category G – Use of MIL-C-22520 tool required for assembly 2/ 6/ 7/	
0013	M17/134-00001 M17/134-00003
0014	M17/134-00002 M17/134-00004
0015	M17/116-RG307
0016 3016	M17/45-RG108 M17/186-00001
0017 3017	M17/176-00002 5/
0018	M17/177-00001 5/
0019	M17/178-00001
0020	M17/179-00001
0021 3021	M24643/33-01UN
0024	M17/135-00003 M17/135-00005
0025	M17/135-00004 M17/135-00006

1/ The latest version of each cable shall be applicable.
2/ These connectors have captivated center contacts.
3/ Not for Navy use.
4/ Inactive for new design (see table III).
5/ Cables to be used for the +200°C thermal shock test.
6/ These connectors are assembled using the applicable crimping tool to the specified cables.
7/ Complete connector assembly shall consist of a body, center contact, intermediate contact, ferrule, and assembly instructions.

Corona level:

Altitude: 70,000 feet.

Voltage: 200 V rms minimum.

RF high potential withstanding voltage:

800 V rms, between center conductor and intermediate conductor.

200 V rms, between intermediate conductor and outer conductor at 5 MHz to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable .200 inch to .325 inch outside dimension). 40 pounds minimum.

Coupling mechanism retention force: Not applicable.

Rise time degradation: 400 picoseconds maximum. (Not applicable to connectors using twin conductor cables.)

Connector durability: 500 cycles minimum at 12 cycles per minute maximum.

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Part or Identifying Number (PIN): M49142/09- (dash number from table I).

Group qualification: See table II.

Cross-reference of cables: See table III.

Retention of qualification: See table IV.

TABLE II. Group qualification. 1/

Groups	Submission and qualification of any of the following dash numbers	Qualifies the following dash numbers
I	*001 *002 *005 *007 *009	*001 *002 *005 *007 *009
II	*003 *004 *010 *011	*003 *004 *010 *011
III	*006 *012	*006 *012
IV	*013 *014 *015 *018	*013 *014 *015 *018
V	*021 *016 *017	*021 *016 *017
VI	*019 *020	*019 *020
VII	*022 *023	*022 *023
VIII	*024 *025	*024 *025

1/ If a connector manufacturer produces a connector which meets all the requirements for two or more connector PIN'S (within same series), the manufacturer may receive qualification approval for two or more connector PINs by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate PIN. For group qualification, the connectors must be of similar design. Qualification of connectors qualifies connectors of the same body material and finish only. * Designates body material and finish.

TABLE III. Cross-reference of cables.

Preferred cable	Superseded cable
M17/177-00001	380-10045-1
M17/178-00001	D3-7619-5/336
M17/179-00001	D3-7619-5/338

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TABLE IV. Retention of qualification.

Subgroup	/3 & /8		/4 & /10	/5 & /9		/6 & /11	
1	/3-0008	---	/4-0004	---	---	---	---
2	/3-0008	/8-0006	/4-0004	---	---	---	/11-0006
3	/3-0008	/8-0006	---	---	---	---	---
4	/3-0008	/8-0006	---	---	---	/6-0007	/11-0006
5	/3-0008	---	/4-0004	---	---	---	---
Units	15	9	9	0	0	3	6

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, AT, MI
Navy - AS, MC, OS, SH
Air Force – 19, 99

Preparing activity:
DLA - CC

(Project 5935-4550-006)

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